

# KARTIK MADAN

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## EXPERIENCE

### Infosys

#### Specialist Programmer III

 November 2020 – Present  London, United Kingdom

#### AI Engineer | BP – Trading Ops Agent

**Problem:** Build a conversational AI platform enabling trading operators to query shipment, parcel, deal, and portfolio data using natural language over BP’s trading warehouse.

- Architected a four-stage **multi-agent Text-to-SQL platform** serving real-time operational queries over BP’s trading warehouse.
- Built a context resolution layer combining business rules, warehouse metadata, and domain terminology to generate accurate and **governed SQL** over enterprise datasets.
- Designed query safety, validation, and **evaluation frameworks** to improve reliability, security, and correctness of AI-generated responses.
- Delivered the **end-to-end platform** across backend services, conversational interfaces, authentication, session management, and automated deployment pipelines.

#### AI Engineer | BP – Contract Value Leakage

**Problem:** Detect and prevent contract value leakage by reconciling supplier invoices against contracted price books.

- Architected a **multi-agent AI platform** combining OCR, information extraction, and anomaly detection to reconcile **~15K supplier invoices/month** against contract price books.
- Built intelligent document-processing pipelines supporting 200+ supplier templates with 95%+ extraction accuracy.
- Developed AI-powered extraction and validation services **across 50+ invoice formats**.
- Implemented invoice-to-contract matching and leakage detection capabilities, reducing undetected contract leakage by **~40%**.

#### Senior Full Stack Engineer / Project Lead | BP – Wild Bean Café

**Problem:** Create an enterprise-grade API and analytics platform to automate pricing, sales, and inventory management for self-service **Micro Market** units across 100+ BP forecourt retailers.

- Led development of a cloud-native platform integrating sales, inventory, and pricing data across BP retail locations in the UK and Poland.
- Built enterprise integration and analytics workflows connecting POS systems, SAP, and cloud data platforms.
- Developed **conversational analytics** capabilities enabling retail managers to query operational data using natural language.
- Contributed to a **148% improvement in dealer profitability** through automation and reporting enhancements.

#### Full Stack Engineer | BP – Store of Today

**Problem:** Automate **event detection** using computer vision and deploy **dynamic pricing** over Electronic Shelf Labels (ESLs) in BP retail stores in Australia.

- Built event-driven computer vision solutions for real-time hazard detection across retail locations.
- Developed dynamic pricing capabilities using Electronic Shelf Labels (ESLs).
- Automated AWS infrastructure and deployment pipelines using Terraform and cloud-native tooling.

#### Full Stack Engineer | Infosys Equinox Studio

- Developed the Infosys Equinox Studio, a **low-code/no-code digital commerce platform** used by global clients including Kraft Heinz, Shutterfly, Herbalife and Nu Skin Enterprises which accounted for **22% of the Equinox team’s total revenue**.

## SKILLS

- AI/ML: LLMs, RAG, Agentic AI, LangChain, TensorFlow, PyTorch
- Languages: Python, Java, Node.js, JavaScript
- Cloud: AWS, Azure, GCP, Terraform
- Data: BigQuery, Spark, Airflow, Athena, SQL/NoSQL
- Tools: Git, Redis, Kafka, Jenkins, Grafana

## PROJECTS AND OPEN SOURCE

**Energy Trading:** ML-driven market participation platform for India’s power trading ecosystem.

**Microsoft Agent Framework:** Active contributor to Microsoft’s framework for building, orchestrating and deploying AI agents and multi-agent workflows.

**ContractIQ:** Developed a system to create and analyze legal contracts using LangChain.

## ACHIEVEMENTS

- **Winner** – Smart India Hackathon 2019
- **Winner** – Barco Geekathon 2018
- Won Infosys RISE Award for Business Ninja in Q3FY24 and Tech Wizard Q4FY23
- **Codeforces** Best Finish: World Rank 625 | National Rank 37 (of 7585 participants)
- **LeetCode Knight** with 1,854 Contest Rating

## EDUCATION

Guru Gobind Singh Indraprastha University  
**B. Tech (Electronics and Comm. Technology) (8.2GPA)**  
 2016 – 2020

## PUBLICATIONS

- **Development of a Real-Time Pollution Monitoring System for Green Auditing**
- **Analyzing time series data for Forex and Stock Market Trends using recurrent neural networks**
- **Patent Application: System and Method for identifying and replacing proprietary images with synthetically generated images**

## COURSEWORK

- Algorithm Design and Analysis
- Data Structures
- Object Oriented Programming
- Database Management System
- Machine Learning
- Computer Networks